

Principal Role: Workflow Documentation

Faculty Management System · Scope: Principal role · Budget, emergency budget and purchase are excluded · Date: 2026-10-09

Method. Written from the actual code (navigation, pages, API permission checks, status values). Tags: **[Verified]** = read in code, **[Partly verified]** = permission checked but page not opened, **[Not verified]** = not checked.

Readers. Developers, testers, UI/UX designers, and anyone new to the system.

1. The Principal in simple words

The Principal is the head of one college. Other people ask for things: the HOD asks to hire someone, a teacher asks for leave. The Principal looks at the request and decides: yes, no, or "fix it first". The system then tells the person who asked.

The Principal also watches the college (students, timetable, marks, attendance) and sets up some rules (settings, who holds which job seat, circulars).

The Vice Principal (VP) shares almost every Principal page and API. The differences are in section 7.

1.1 Who is treated as Principal

Login	What happens
Principal	Full powers [Verified]
Vice Principal	Same pages, plus its own home page and "General Admin Vacancies" [Verified]
College Admin, Director	Turned into Principal at sign-in so the pages open, but the API refuses their decisions (hiring, leave, interview plan, candidates) with 403. Their menu hides My Profile, My Attendance, My Leave, Hiring Requests and Leave Approvals [Verified]

2. Login to logout journey

1. Open the login page.
2. Type email and password. Firebase checks them.
3. The system calls `POST /api/auth/session` . It verifies the token, finds the role and college, and sets a signed cookie `fms-session` valid for 24 hours.
4. The browser opens `/principal` (or the page named in `?redirect=`).
5. A page gate (`proxy.ts`) lets this role open only `/principal/*` and a few shared paths: panel interviews, evaluation, candidate profile, leave adjustments.
6. The dashboard appears.
7. The Principal works through the menu. Every action is checked again on the server.
8. Logout clears the session cookie.

Special case: a Principal linked to a faculty record whose status is RESIGNED or RETIRED can still sign in but becomes read-only. Every write is refused.

3. Dashboard audit (`/principal`)

The screen shows "Hello, <first name>", three summary cards and two lists.

Card	Shows	Click opens
Pending Vacancies	Vacancy requests with status PENDING	<code>/principal/vacancies</code>
Interviews & Decisions	Hiring batches with status PENDING	<code>/principal/interviews</code>
Departments	Top-level departments	<code>/principal/courses?tab=departments</code>

Lists: **Pending Hiring Requests** (View all, Review) and **Pending Interview Plans** (View all, Review).

Loading state: cards show a dash and lists show skeleton placeholders. If the Super Admin hides a module, the card link falls back to a visible page.

Gap: there is no card for leave approvals, attendance or students, although leave is a frequent Principal task.

4. Sidebar and navigation

The Super Admin can hide any item per college. Hiding is cosmetic only. Every action has its own server check.

Section	Items and routes
Top	Dashboard <code>/principal</code>
Academics	Courses <code>/principal/courses</code> ; Students <code>/principal/students</code> (tabs Roster, Promotion, Graduated); Timetable View <code>/principal/timetable</code> ; Internal Marks <code>/principal/internal-marks</code>
Staff & HR	Faculty <code>/principal/faculty</code> ; Staff <code>/principal/staff</code> ; Circulars <code>/principal/circulars</code>
Leave & Attendance	Leave Approvals <code>/principal/leave-approvals</code> ; Leave History <code>/principal/leave-history</code> ; Attendance <code>/principal/attendance-report</code> ; Attendance Reports <code>/principal/attendance-reports</code>
Hiring Pipeline	Hiring Requests <code>/principal/vacancies</code> . Later steps reached from it: Interviews <code>/principal/interviews</code> , Negotiate <code>/principal/negotiate/[id]</code> , Decisions <code>/principal/decisions/[id]</code> , Appointment Letters <code>/principal/appointment-letters</code>
Personal	My Profile, My Attendance, My Leave
Administration	Audit Logs, Role Assignments, Settings

Old routes still work but are no longer in the menu: `/principal/promotions` , `/graduates` , `/attendance-completion` , `/absent-report` , `/shortage-report` , `/faculty-not-posted` , `/attendance-history` . They exist for bookmarks and notification links.

5. Permission matrix

V = view, C = create, E = edit, A = approve, R = reject, D = delete. "API only" means the server would accept the call but the screen has no button.

Area	Principal / VP	Others	Notes
Hiring request (vacancy)	V, A, R	HOD creates. College Admin: V only	Decision is final
Interview plan (batch)	V, A, R, Modify	HOD submits and sets up. College Office adds venue and documents	College Admin refused
Candidate decision	A, R, set negotiation terms	Panel members score	College Admin refused. Final
Appointment letter	V, C	Super Admin creates. College Office, HOD, Accounts read	
Offer letter	V, C	College Office, Accounts can create	
Leave, final stage	A, R	College Admin refused	Leave routed to the Principal can only be decided by the Principal
Leave, HOD stage	A, R only if HOD stage is stuck	HOD decides normally	
Circulars	V, C, E, publish	Others only if the Principal or VP grants permission	No approval step
Faculty	V, C, E, D	HOD same. College Office V only	
Staff and users	V, C, E; D	HOD, College Office V, C, E	Delete: Principal and VP only
Departments	V, C, E, D	HOD, Super Admin	
Courses	V; C for Principal only	Super Admin creates	VP cannot create (likely oversight)
Sections	V	Create: HOD, Super Admin	
Students	V; Promotion tab promotes		API only: add, edit, delete, bulk-delete
Timetable	V (published)		API only: create slots, publish, reset
Internal marks	V	Faculty enters marks	
Attendance	V, import, manual entry		
Audit logs	V (latest 100)		
Role seats	Assign, vacate, update, remove, reactivate	Super Admin, Management, Administration	
Settings	V, E	HOD may read Faculty Norms	
Student permission routing	Principal V, E; VP V only	HOD E for own department	

The Principal must not: decide anything as College Admin; create sections; enter internal marks; create exam circulars (Exam Cell only); approve their own leave (it goes to Management); see another college's data.

6. Module-by-module audit

6.1 Hiring Requests (vacancies)

- **Purpose.** The HOD asks to fill a post; the Principal decides if it is allowed.
- **Entry.** Dashboard card or list, or sidebar Hiring Requests (`/principal/vacancies`).
- **View.** Pipeline board by department and status. Each request shows position, department, required vs available count, student strength, cadre ratio data, HOD justification, hiring mode.
- **Actions.** Approve (`/vacancies/[id]/approve`), Reject (`/vacancies/[id]/reject`).
- **Validation.** Status is required and must be APPROVED, REJECTED or MODIFIED (check added in this audit). The reject reason is optional.
- **Confirmation.** Each decision has its own page with a Cancel button. No second dialog.
- **Result.** Decision, decider, time, reason and notes are stored. The HOD and Department Office head are notified. An audit entry is written.
- **Next step.** If approved, the HOD collects candidates. If rejected, it ends.
- **Errors.** A request already APPROVED or REJECTED cannot change to a different decision (409). The page now shows that message. A missing request returns 404.
- **Permission.** Principal and VP only. College Admin gets 403.

6.2 Interviews (interview plans)

- **Purpose.** Approve the plan for interviewing candidates.
- **Entry.** Interviews card (`/principal/interviews`), then `/interviews/[id]` .
- **View.** Batches with position, department, candidates, panel, date and time, hiring mode.
- **Actions.** Approve, Reject, Modify, with Principal notes.
- **Validation.** Only Principal or VP, and only while the batch waits for the Principal. College Admin gets 403. An unknown phase value is refused (400).
- **Result.** Approve moves the batch to HOD_FINAL_SETUP and tells the HOD to add venue, required documents, demo classroom and coordinator. Reject and Modify tell the HOD with the notes.
- **Next step.** HOD completes setup, interviews run, then the batch returns to the Principal.

Batch phases in order: PRINCIPAL_REVIEW, HOD_FINAL_SETUP, INTERVIEW_READY, IN_PROGRESS, PANEL_INTERVIEW, PRINCIPAL_FINAL_REVIEW, COMPLETED. The HOD may only move phases forward one step at a time.

6.3 Negotiate and Decisions (candidate hiring decision)

- **Purpose.** Agree salary and joining date, then hire or not.
- **Entry.** When a batch reaches PRINCIPAL_FINAL_REVIEW all Principals are notified. Open `/principal/negotiate/[id]` , then `/principal/decisions/[id]` .
- **Negotiate.** Per candidate: expected salary, negotiated salary, date of joining, terms and conditions. Save.
- **Decide.** Approve or Reject each candidate.
- **Validation.** Approval needs a negotiated salary above zero and a joining date not in the past. Only Principal and VP can write these fields; others get 403.
- **Result.** Candidate status becomes APPROVED or REJECTED. The vacancy count updates and the batch owner is notified. When every candidate is decided the batch becomes COMPLETED.

- **Finality.** A decided candidate cannot be flipped (409).
- **Next step.** The Principal issues the Appointment Letter. The candidate accepts or rejects on the public offer page. College Office handles documents and joining.

6.4 Appointment Letters

Entry `/principal/appointment-letters` . Create a letter with CTC, address and terms; status becomes SENT. Principal, VP and Super Admin create. College Office, HOD and Accounts can read.

6.5 Leave Approvals and Leave History

- **Purpose.** Final decision on staff leave.
- **Entry.** `/principal/leave-approvals` .
- **What reaches the Principal.** (a) An "Other" leave forwarded by the HOD, status PENDING_VICE_PRINCIPAL: Principal or VP can decide. (b) A leave the applicant routed to the Principal, status PENDING_PRINCIPAL: Principal only. (c) Leave from staff with no HOD stage. (d) An HOD-stage leave when no HOD can act.
- **Actions.** Approve or Reject, with remarks.
- **Validation.** Approving an "Other" leave requires a category: Maternity, Family Planning, Quarantine, Extraordinary or Compensatory. Maternity needs female staff with at least one year of service. A VP cannot approve their own leave. College Admin gets 403.
- **Result.** Approve commits the balance (days over the balance become Loss of Pay). Reject releases the reserved balance. Both run in one transaction, so a double click cannot deduct twice. The applicant and substitutes are notified; approved leave is copied into attendance.
- **Other actions on the same route.** Cancel, edit, propose coverage, verify or reject On Duty proof and certificates.
- **Leave History** (`/principal/leave-history`). By department, then person, then type. Read only.
- **The Principal's own leave** goes to Management, not here.

6.6 Faculty

Entry `/principal/faculty` , then department, then member. View department and faculty lists and profile modules (personal, academic, teaching load, resume, previous teaching assignments). Actions: Add Faculty, Import (Excel), edit sections, manage credentials, delete a faculty record, supporting staff. Saving a member as RESIGNED or RETIRED vacates their seats and makes their login read-only. Risk: delete has only a confirmation dialog, no approval.

6.7 Staff and users

Entry `/principal/staff` . View non-faculty staff and modules; add, edit, credentials. Deleting a user is Principal and VP only.

6.8 Courses and Departments

Entry `/principal/courses` (Courses and Departments tabs). Create, edit and import departments; add courses; set course length, years taught and timings; edit the course catalog. Empty Years Taught means "not configured", not "all years". Editing Years Taught is refused (409) while sections, students, assignments, timetable slots or exam setups exist for a year being removed. A course cannot be shortened while later years hold data.

6.9 Students

Entry `/principal/students` , tabs Roster, Promotion, Graduated. Roster: list and strength dashboard, filters for course, department, year, section and search; click a row for the student detail. Promotion promotes students to the next year. Graduated lists graduates. By design the roster has no Add, Edit, Delete, Import or Export button. Mismatch: the API still lets Principal and VP add, edit, delete, bulk-delete and promote.

6.10 Timetable View and Internal Marks

Timetable View shows published timetables by course-year and section, read only on screen (the API would also let the Principal create slots, publish and reset). Internal Marks lists marks by assignment with a read-only detail dialog; no edit or approval.

6.11 Attendance and Attendance Reports

Attendance (`/principal/attendance-report`): daily staff check-in and check-out, import, manual entry. Attendance Reports (`/principal/attendance-reports`): reports, history, completion, absent, shortage, faculty not posted, student attendance. Filters: date, department, course, section. The completion view redirects a College Admin login away.

6.12 Circulars

Entry `/principal/circulars` . Status values DRAFT, PUBLISHED, ARCHIVED. Actions: create, edit, publish, archive; set who may send (Circular Permissions) and circular settings. The Principal and VP can always send. No approval step exists.

6.13 Role Assignments (seats)

Entry `/principal/role-assignments` . Seats: Principal, College Admin, Vice Principal, Academics, HOD (one per department), IQAC Coordinator, T&P, R&D, R&D Coordinator (per department), Placement, Exam Cell, Library, Webmaster. Actions: Assign, Vacate, Update, Remove, Reactivate. A college has exactly one Principal, one VP and one College Admin. Resigned or retired people are not offered and cannot be given a seat.

6.14 Settings

Cards on `/principal/settings` : College Information, Faculty Norms, Student Permissions, Leave Module, Leave Approval Routing, Vacation Staff, Leave Type Rules. Each Save takes effect at once, with no approval step.

6.15 Audit Logs

`/principal/audit-logs` shows the latest 100 events for the college, newest first. No filter, search, export or paging. Read only.

6.16 My Profile, My Attendance, My Leave

The Principal's own records. Hidden for College Admin. Own leave goes to Management.

7. Principal compared with Vice Principal

- Same pages and almost the same permissions.
- A leave routed to the Principal can only be decided by the Principal; one routed to the VP can be decided by either.
- A VP cannot approve their own leave.
- The VP has "General Admin Vacancies" and "Adjustment Requests" (accept or decline being a substitute). The Principal is never named as a substitute.
- The VP can read but cannot save the student permission routing.
- The VP cannot create a course.

8. Workflow diagrams

8.1 Hiring: vacancy to appointment letter

Step	Who	What happens	Alternate path
1	HOD	Submits vacancy request (PENDING)	
2	Principal	Reviews the request	Approve: go to 3. Reject: HOD told with reason, END
3	HOD	Collects candidates and submits an interview plan (PRINCIPAL_REVIEW)	
4	Principal	Reviews the plan	Approve: HOD_FINAL_SETUP. Modify: HOD fixes and resubmits. Reject: END
5	HOD, College Office	Venue, documents, demo room, coordinator (INTERVIEW_READY)	
6	Panel	Demo class and interviews (IN_PROGRESS, PANEL_INTERVIEW)	
7	HOD	Sends batch back (PRINCIPAL_FINAL_REVIEW); Principals notified	
8	Principal	Enters salary, joining date, terms (Negotiate)	
9	Principal	Decides each candidate (Decisions)	Approve needs salary above zero and future joining date. Reject: candidate ends
10	System	All decided: batch COMPLETED	
11	Principal	Issues Appointment Letter (SENT)	
12	Candidate, College Office	Candidate accepts or rejects; Office handles joining	END

8.2 Leave

Step	Who	What happens	Alternate path
1	Staff member	Applies for leave	If applicant is the Principal: goes to Management. If routed to Principal or no HOD stage: starts at step 3

Step	Who	What happens	Alternate path
2	HOD	Reviews	Standard type approve: balance committed, APPROVED, END. Reject: balance released, REJECTED, END. "Other" type: tags paid/unpaid and forwards
3	Principal or VP	Reviews forwarded request	Approve (with Other category) or Reject
4	System	Notifies applicant and substitutes; copies approved leave into attendance	END

8.3 Role seat

1. Principal opens Role Assignments and picks a seat and an action.
2. The system checks the actor may do it and the person is eligible (not resigned or retired).
3. The seat is saved; the person's role changes at the next session refresh. END.

8.4 Circular

DRAFT, then PUBLISHED (by anyone allowed to send, including the Principal and VP), then optionally ARCHIVED. END.

8.5 Promotion

Students, Promotion tab, choose the group and confirm. Students move to the next year; final-year students become Graduated. END.

9. Relationship with other roles

Role	Who starts	Principal's part	Where it ends
HOD	Vacancy request, interview plan, Other-leave forward	Decides, HOD is notified	HOD proceeds, or the request ends
Vice Principal	Shared work	Same authority, see section 7	
Faculty	Leave application	Views profile, attendance, history; edits records; decides Other leave	Leave APPROVED or REJECTED
College Office	Data entry, offers, venue, documents	Decides; Office finishes joining	Candidate joins
Students		View and promote only	Students never contact the Principal
Management		Decides the Principal's own leave	
Super Admin		Controls which menu items the Principal sees	
Exam Cell, Academics, IQAC, Library		Principal places them in seats	
Dean		No Principal workflow found	

10. Notifications and audit trail

- Every decision writes an audit entry (who, when, what) and notifies the person who asked.
- Vacancy decisions notify the department's HOD and Department Office head.
- Interview plan decisions notify the department heads.
- Leave decisions notify the applicant and substitutes.
- Reaching Principal final review notifies all Principals.
- The Principal can read the latest 100 audit entries.

11. Status values found in code

Item	Values
Vacancy	PENDING, APPROVED, REJECTED, MODIFIED
Hiring batch phase	PRINCIPAL_REVIEW, HOD_FINAL_SETUP, INTERVIEW_READY, IN_PROGRESS, PANEL_INTERVIEW, PRINCIPAL_FINAL_REVIEW, COMPLETED
Candidate stage	DEMO, INTERVIEW, SALARY_NEGOTIATION, DECISION
Candidate status	PENDING, ARRIVED, SHORTLISTED, APPROVED, REJECTED
Leave	PENDING_HOD, PENDING_PRINCIPAL, PENDING_VICE_PRINCIPAL, PENDING_MANAGEMENT, APPROVED, REJECTED, CANCELLED
Circular	DRAFT, PUBLISHED, ARCHIVED
Offer letter	DRAFT, SENT, ACCEPTED, REJECTED
Seat actions	ASSIGN, VACATE, UPDATE, REMOVE, REACTIVATE

12. Interaction test results

Check	Result
Code review of permission checks (vacancy, interview plan, candidate, leave, circular, seats, faculty, students, timetable, settings, student permission routing)	Pass, except the problems in section 13
Route guard test <code>src/app/api/routeGuards.inventory.test.ts</code>	Pending: still running when this was written
Typecheck <code>tsc --noEmit</code> after the edits	Pending: still running
Clicking each button in a browser, direct URL access, back navigation, real API calls against a database	Not tested. Needs a seeded non-production Firebase environment (<code>tests/e2e/.env.test</code>), not confirmed available
Playwright end-to-end specs for Principal screens	Not run

Nothing was run against real institutional data.

13. Problems found

High

1. **Students screen vs API.** The page has no Add, Edit or Delete, but the API lets Principal and VP add, edit, delete, bulk-delete and promote students.
2. **Timetable screen vs API.** The page is read only, but the API lets the Principal create slots, publish and reset a timetable.

Medium

1. **Vacancy status was not validated.** The route stored any text. **Fixed.**
2. **Hidden error messages** on the vacancy approve and reject pages. **Fixed.**
3. **Reject reason is optional** for a hiring request, so the HOD may get no explanation.
4. **No Forward or Reopen.** A mistaken vacancy or candidate decision cannot be undone from the UI.
5. **Destructive actions without approval:** deleting faculty, users, departments, academic years and sessions.
6. **Course creation excludes the VP,** though the VP can create departments.
7. **Dashboard shows hiring only,** not leave, attendance or students.

Low

1. Circulars have no approval step and many roles can publish.
2. Audit logs are capped at 100 rows with no filter, search, export or paging.
3. A leave routed only to the Principal has no documented delegate when the Principal is away.
4. The Dean role has no defined Principal workflow.
5. College Admin sees pages whose decisions return 403; check each shows a clear message.
6. The same vacancy decision sent twice is accepted again and re-sends notifications.
7. Only Principal and HOD can save student permission routing; confirm the VP read-only intent.

Checked and cleared

- Student permission routing is guarded (Principal, VP, HOD can read; Principal and HOD can save).
- The interview plan route does block College Admin from deciding.
- Vacancy, candidate and leave decisions refuse a changed final decision (409).

14. Fixes made

File	Change
src/app/api/college/vacancy-requests/[id]/route.ts	PATCH returns 400 unless status is APPROVED, REJECTED or MODIFIED
src/app/(dashboard)/principal/vacancies/[id]/approve/page.tsx	Shows the server's own error message instead of a generic one
src/app/(dashboard)/principal/vacancies/[id]/reject/page.tsx	Same

Items 1, 2 and 5 to 16 were left unchanged on purpose: they are product decisions or wider changes.

15. Remaining and next steps

- Confirm the route guard test and the typecheck results.
- Set up the seeded test environment and run the end-to-end specs for the Principal screens.
- Decide items 1 and 2 (students and timetable), then align screen and API.
- Decide whether a reject reason should be required and whether a decision can be reopened.
- Consider leave, attendance and student cards on the dashboard.

16. One complete journey

Login → Dashboard → Pending Hiring Requests → Review → Approve or Reject → HOD notified → Interviews → open batch → Approve, Modify or Reject → HOD sets up → Negotiate → salary and joining date → Decisions → Approve or Reject candidate → Appointment Letter → Leave Approvals → open request → Approve or Reject → applicant notified → Browse Students, Timetable, Marks, Attendance Reports → Faculty and Staff changes → Circulars → Role Assignments → Settings → Audit Logs → Logout.